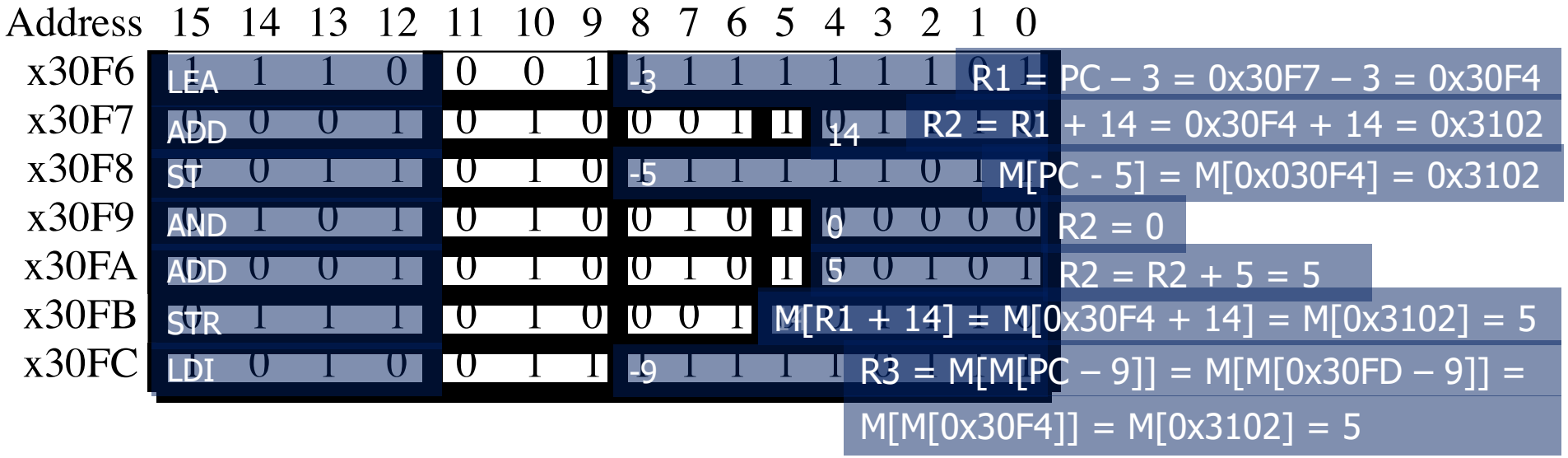
Assembly Programming

Readings

- *Introduction to Computing Systems: From Bits and Gates to C and Beyond by* Patt & Patel
  - Chapter 5, 6
  - Appendix A – LC3 ISA

Addressing Example in LC-3

- What is the final value of R3?



- The final value of R3 is 5

Control Flow Instructions

- In LC-3 and in MIPS there are three types of opcodes
  - Operate
  - Data movement
  - Control

Control Flow Instructions

- Allow a program to execute out of sequence
- Conditional branches and jumps
  - Conditional branches are used to make decisions
    - E.g., if-else statement
  - In LC-3, three condition codes are used
  - Jumps are used to implement
    - Loops
    - Function calls
  - JMP in LC-3 and j in MIPS

Condition Codes in LC-3

- Each time one GPR (R0-R7) is written, three single-bit registers are updated
- Each of these condition codes are either set (set to 1) or cleared (set to 0)
  - If the written value is negative
    - N is set, Z and P are cleared
  - If the written value is zero
    - Z is set, N and P are cleared
  - If the written value is positive
    - P is set, N and P are cleared
- SPARC and x86 are examples of ISAs that use condition codes

Conditional Branches in LC-3

- BRz (Branch if Zero)

```
BRz   PCoffset9
```

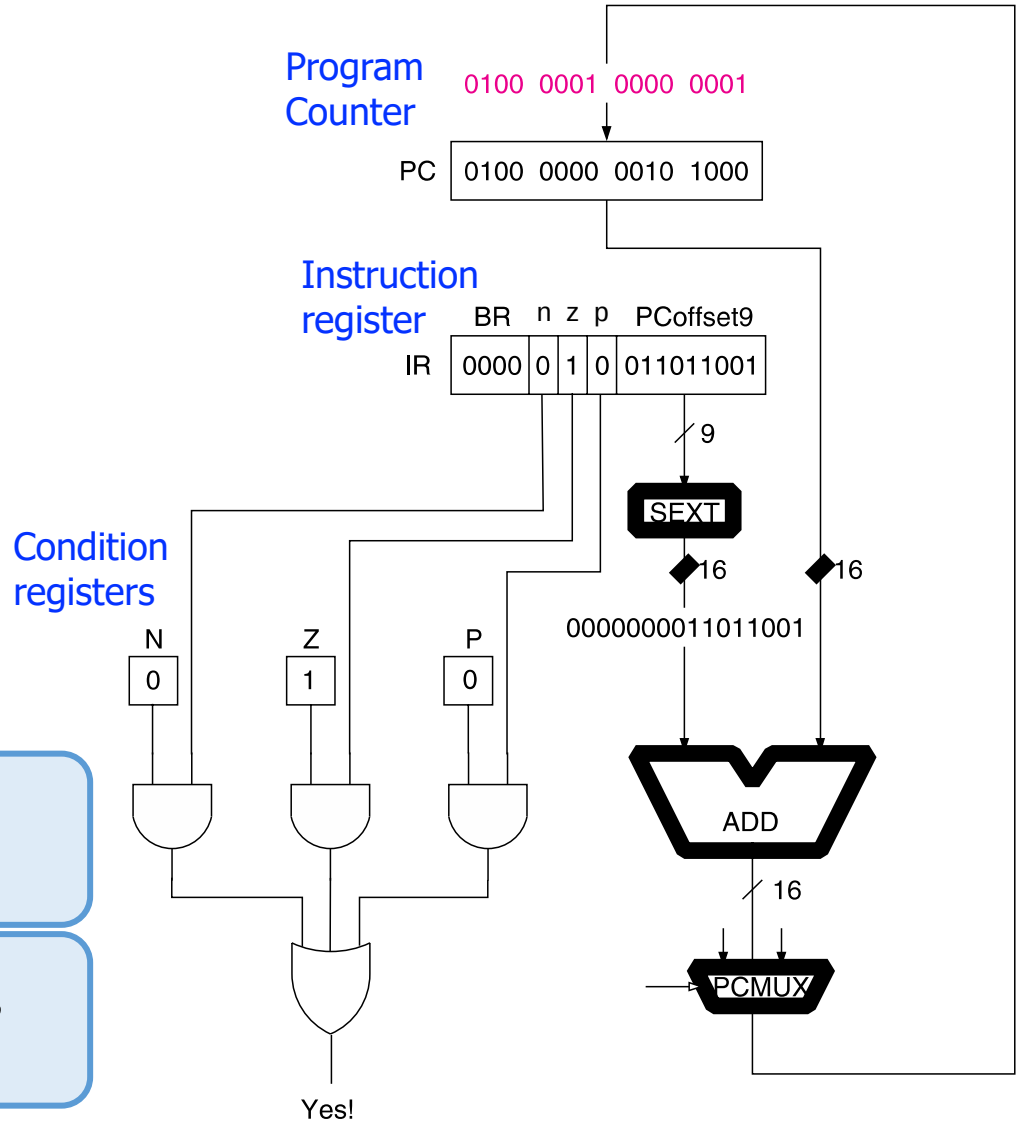
  - n, z, p = which condition code is tested (N, Z, and/or P)
    - n, z, p: instruction bits to identify the condition codes to be tested
    - N, Z, P: values of the corresponding condition codes
  - PCoffset9 = immediate or constant value
  - if ((n AND N) OR (p AND P) OR (z AND Z))
    - then PC ← PC† + sign-extend(PCoffset9)
  - Variations: BRn, BRz, BRp, BRzp, BRnp, BRnz, BRnzp

† This is the incremented PC

Conditional Branches in LC-3

- BRz

```
BRz   0x0D9
```



What if n = z = p = 1?*
(i.e., BRnzp)

And what if n = z = p = 0?

*n, z, p are the instruction bits to identify the condition codes to be tested

Branch If Equal in MIPS and LC-3

MIPS assembly

```
beq  $s0, $s1, offset
```

LC-3 assembly

```
NOT  R2, R1
ADD  R3, R2, #1
ADD  R4, R3, R0
BRz  offset
```

Subtract (R0 - R1)

- This is an example of tradeoff in the instruction set
  - The same functionality requires more instructions in LC-3
  - But, the control logic requires more complexity in MIPS

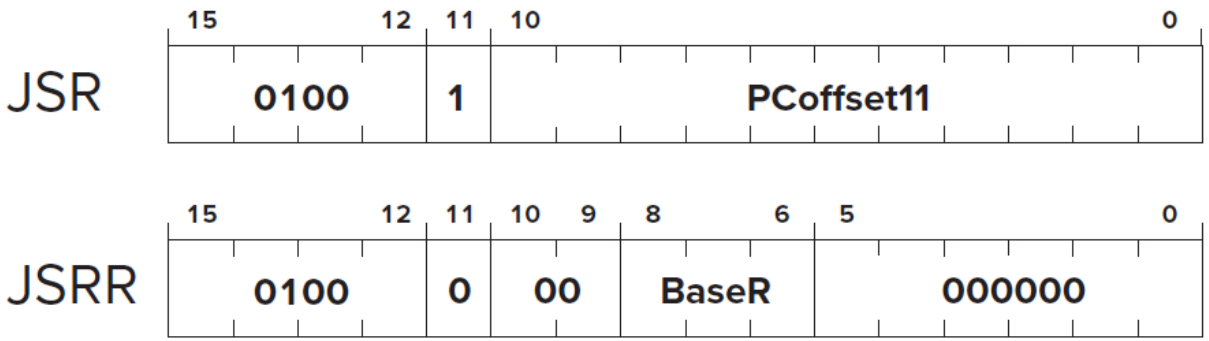
JSR
JSRR

Jump to Subroutine

Assembler Formats

```
JSR    LABEL
JSRR   BaseR
```

Encoding



Operation

```
TEMP = PC;†
if (bit[11] == 0)
   PC = BaseR;
else
   PC = PC† + SEXT(PCoffset11);
R7 = TEMP;
```

JMP
RET

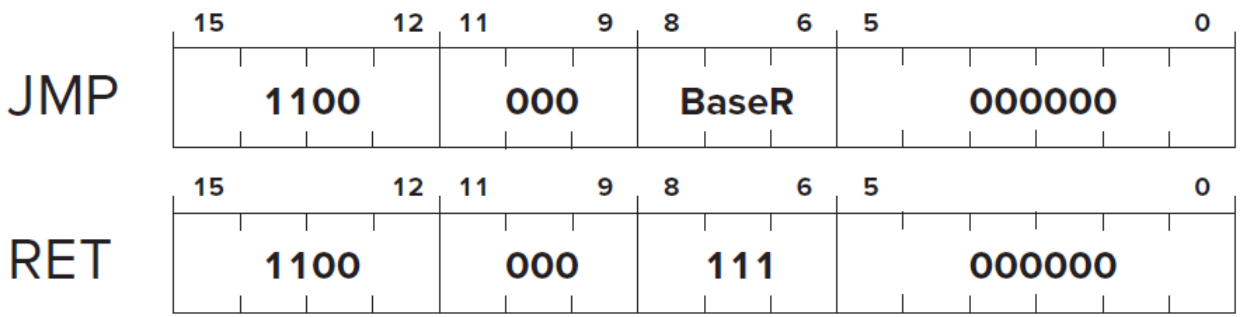
Jump

Return from Subroutine

Assembler Formats

```
JMP   BaseR
RET
```

Encoding



Operation

```
PC = BaseR;
```

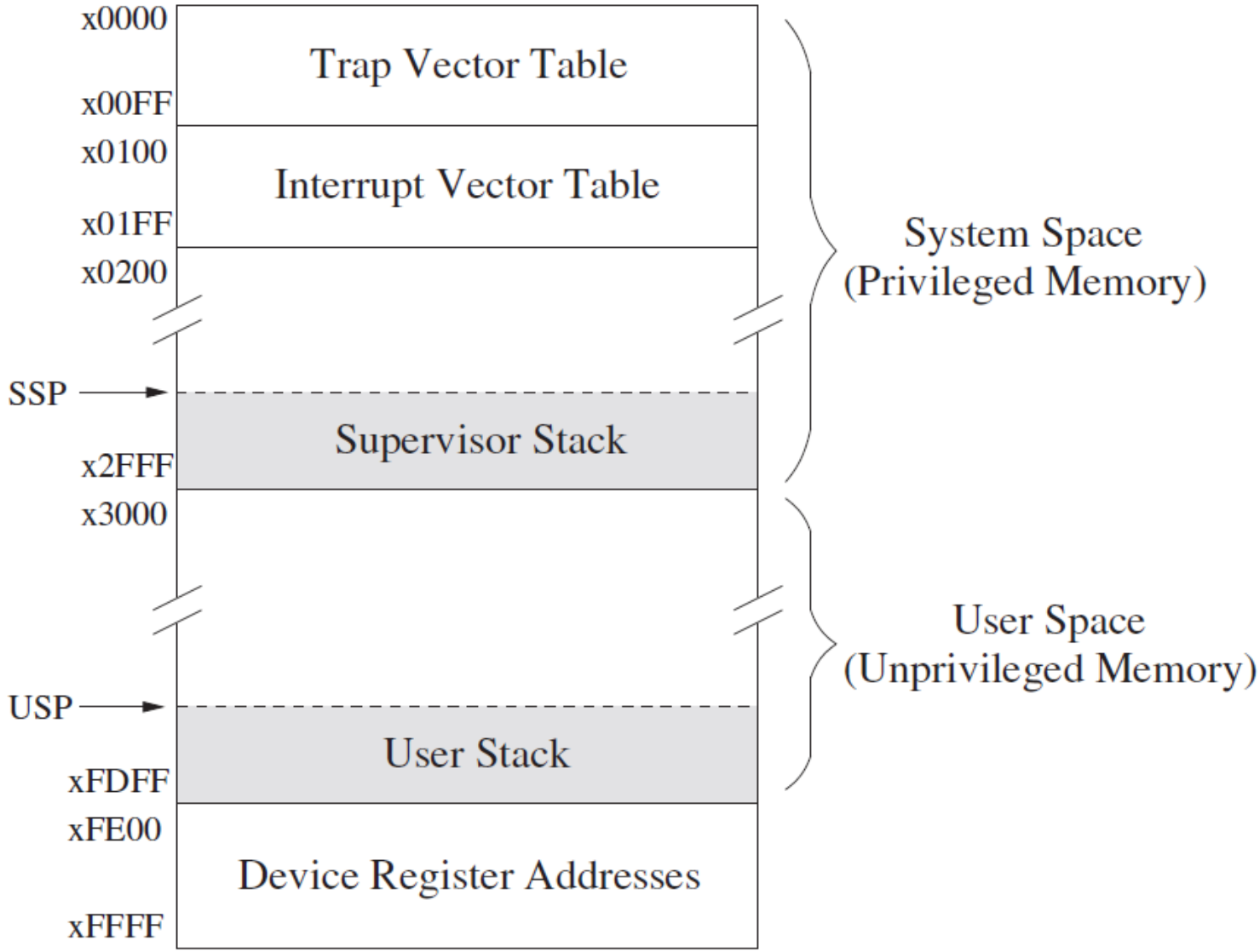


Figure A.1 Memory map of the LC-3

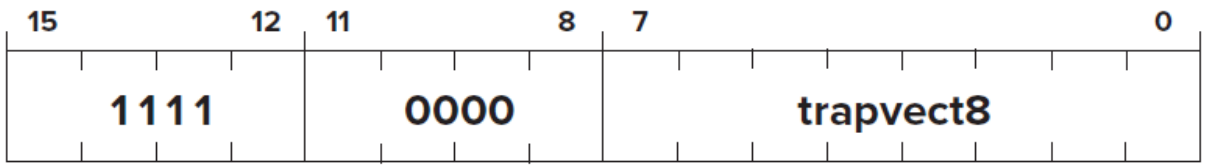
TRAP

System Call

Assembler Format

TRAP trapvector8

Encoding



Operation

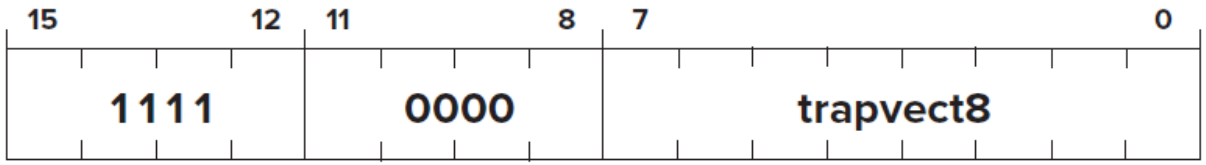
```
TEMP=PSR;
if (PSR[15] == 1)
   Saved_USP=R6 and R6=Saved_SSP;
   PSR[15]=0;
Push TEMP,PC† on the system stack
PC=mem[ZEXT(trapvect8)];
```

TRAP

Assembler Format

TRAP trapvector8

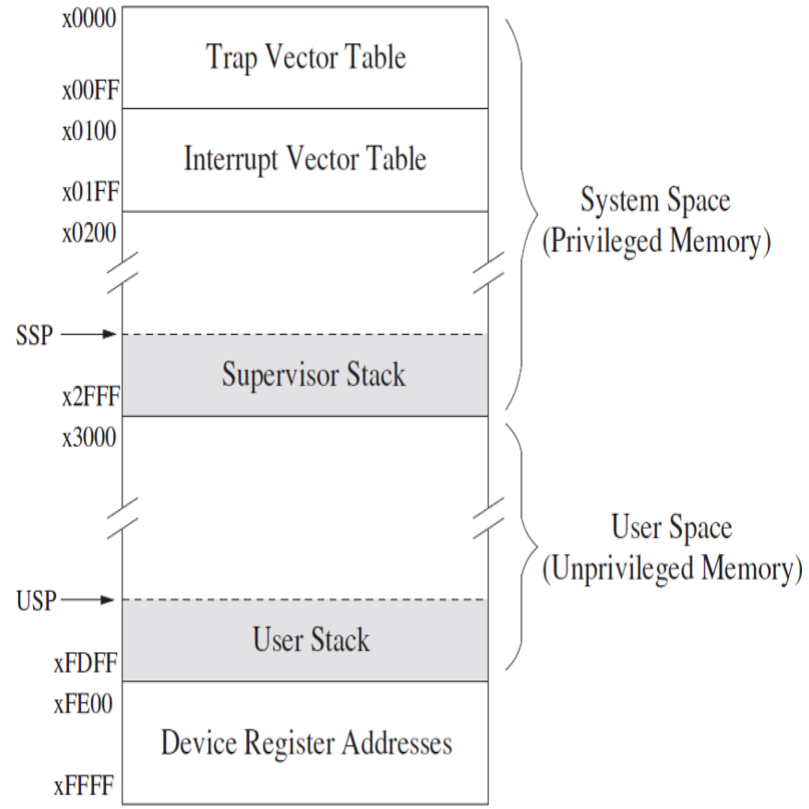
Encoding



Operation

```
TEMP=PSR;
if (PSR[15] == 1)
   Saved_USP=R6 and R6=Saved_SSP;
   PSR[15]=0;
Push TEMP,PC† on the system stack
PC=mem[ZEXT(trapvect8)];
```

System Call



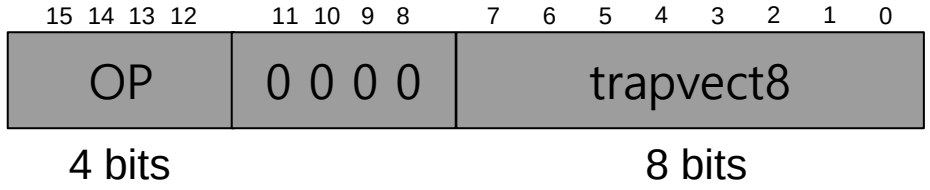
TRAP Instruction

■TRAP invokes an OS service call

LC-3 assembly

```
TRAP 0x23;
```

Machine Code



❑OP = 1111

❑trapvect8 = service call

- 0x23 = Input a character from the keyboard
- 0x21 = Output a character to the monitor
- 0x25 = Halt the program

Table A.3 Trap Service Routines

Trap Vector	Assembler Name	Description
x20	GETC	Read a single character from the keyboard. The character is not echoed onto the console. Its ASCII code is copied into R0. The high eight bits of R0 are cleared.
x21	OUT	Write a character in R0[7:0] to the console display.
x22	PUTS	Write a string of ASCII characters to the console display. The characters are contained in consecutive memory locations, one character per memory location, starting with the address specified in R0. Writing terminates with the occurrence of x0000 in a memory location.
x23	IN	Print a prompt on the screen and read a single character from the keyboard. The character is echoed onto the console monitor, and its ASCII code is copied into R0. The high eight bits of R0 are cleared.
x24	PUTSP	Write a string of ASCII characters to the console. The characters are contained in consecutive memory locations, two characters per memory location, starting with the address specified in R0. The ASCII code contained in bits [7:0] of a memory location is written to the console first. Then the ASCII code contained in bits [15:8] of that memory location is written to the console. (A character string consisting of an odd number of characters to be written will have x00 in bits [15:8] of the memory location containing the last character to be written.) Writing terminates with the occurrence of x0000 in a memory location.
x25	HALT	Halt execution and print a message on the console.

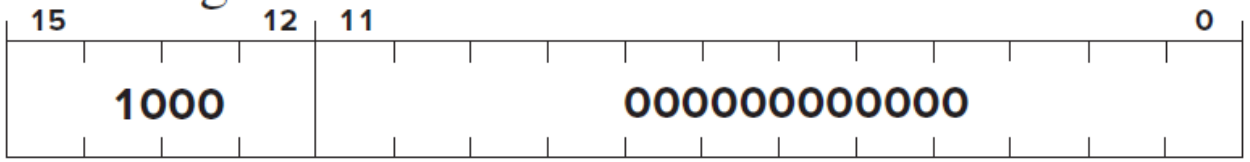
RTI

Return from Trap or Interrupt

Assembler Format

```
RTI
```

Encoding



Operation

```
if (PSR[15] == 1)
   Initiate a privilege mode exception;
else
   PC=mem[R6]; R6 is the SSP, PC is restored
   R6=R6+1;
   TEMP=mem[R6];
   R6=R6+1; system stack completes POP before saved PSR is restored
   PSR=TEMP; PSR is restored
   if (PSR[15] == 1)
      Saved_SSP=R6 and R6=Saved_USP;
```

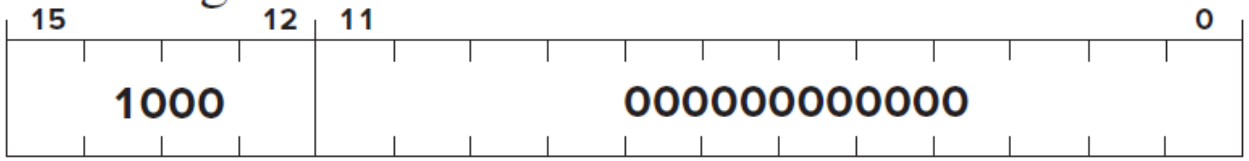
RTI

Return from Trap or Interrupt

Assembler Format

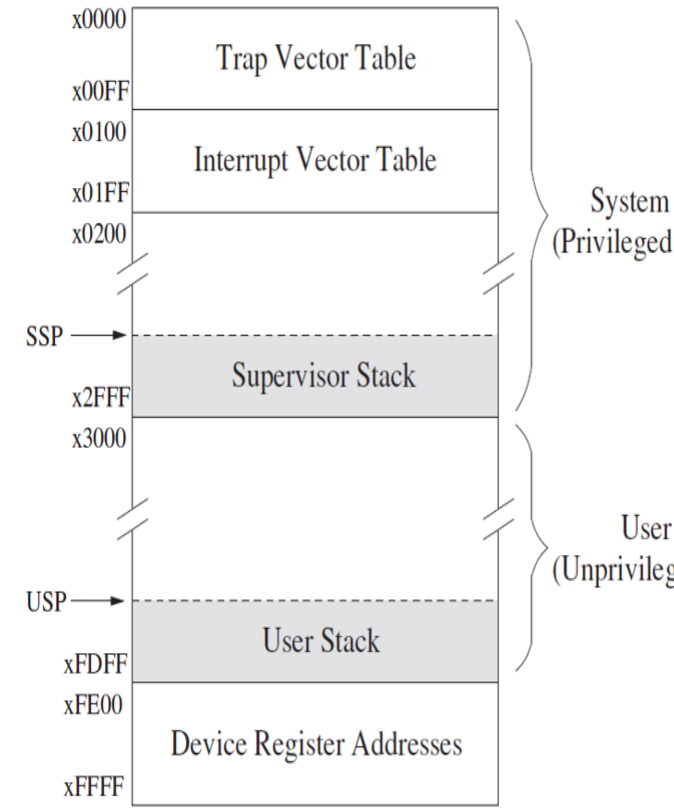
RTI

Encoding



Operation

```
if (PSR[15] == 1)
   Initiate a privilege mode exception;
else
   PC=mem[R6]; R6 is the SSP, PC is restored
   R6=R6+1;
   TEMP=mem[R6];
   R6=R6+1; system stack completes POP before saved PSR is restored
   PSR=TEMP; PSR is restored
   if (PSR[15] == 1)
      Saved_SSP=R6 and R6=Saved_USP;
```



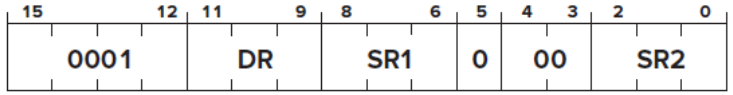
ADD

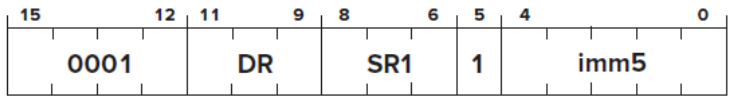
Addition

Assembler Formats

```
ADD   DR, SR1, SR2
ADD   DR, SR1, imm5
```

Encodings





Operation

```
if (bit[5] == 0)
      DR=SR1+SR2;
else
      DR=SR1+SEXT(imm5);
setcc();
```

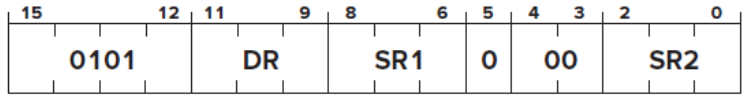
AND

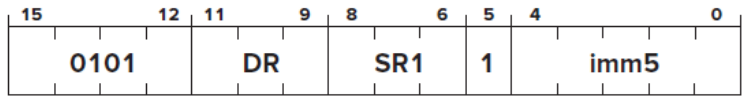
Bit-wise Logical AND

Assembler Formats

```
AND   DR, SR1, SR2
AND   DR, SR1, imm5
```

Encodings





Operation

```
if (bit[5] == 0)
      DR=SR1 AND SR2;
else
      DR=SR1 AND SEXT(imm5);
setcc();
```

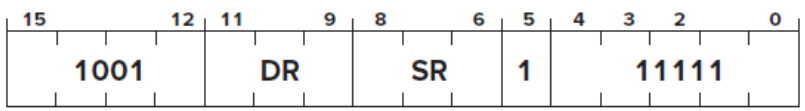
NOT

Bit-Wise Complement

Assembler Format

```
NOT   DR, SR
```

Encoding



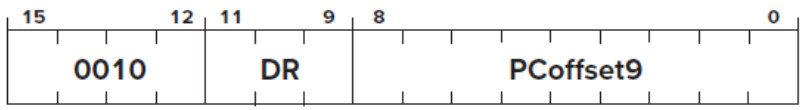
LD

Load

Assembler Format

LD DR, LABEL

Encoding



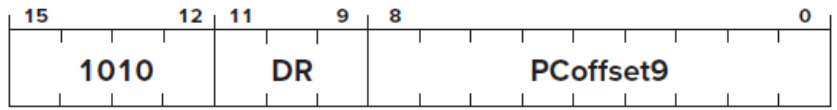
LDI

Load Indirect

Assembler Format

LDI DR, LABEL

Encoding



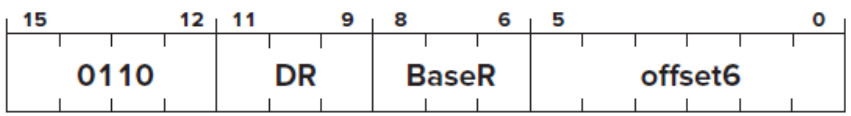
LDR

Load Base+offset

Assembler Format

LDR DR, BaseR, offset6

Encoding



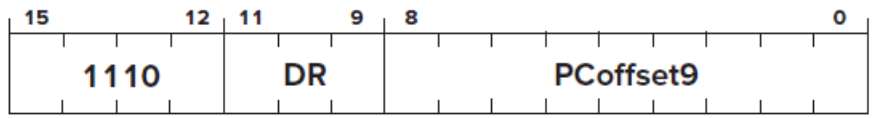
LEA

Load Effective Address

Assembler Format

LEA DR, LABEL

Encoding



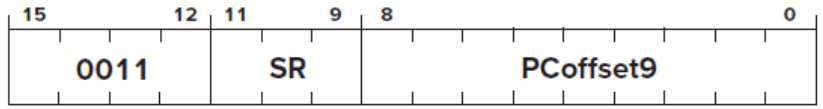
ST

Store

Assembler Format

```
ST   SR, LABEL
```

Encoding



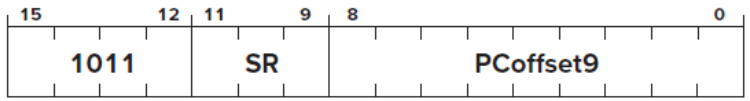
STI

Store Indirect

Assembler Format

```
STI   SR, LABEL
```

Encoding



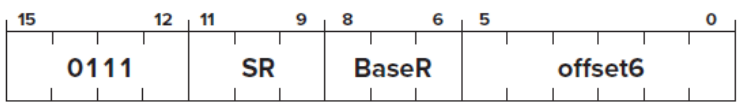
STR

Store Base+offset

Assembler Format

```
STR   SR, BaseR, offset6
```

Encoding



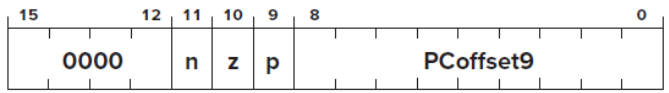
BR

Conditional Branch

Assembler Formats

BRn	LABEL	BRzp	LABEL
BRz	LABEL	BRnp	LABEL
BRp	LABEL	BRnz	LABEL
BR†	LABEL	BRnzp	LABEL

Encoding



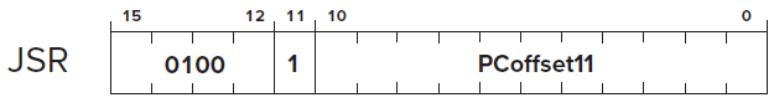
JSR
JSRR

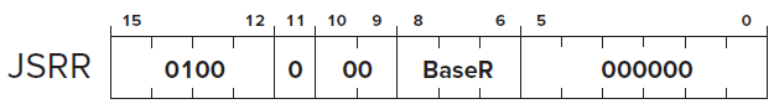
Jump to Subroutine

Assembler Formats

JSR	LABEL
JSRR	BaseR

Encoding





JMP
RET

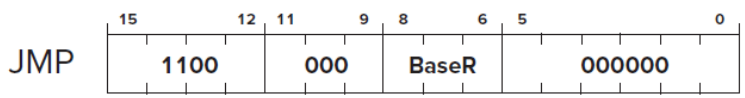
Jump

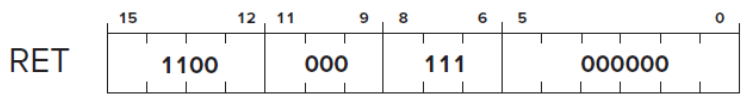
Return from Subroutine

Assembler Formats

JMP	BaseR
RET	

Encoding





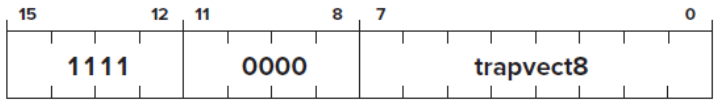
TRAP

System Call

Assembler Format

TRAP trapvector8

Encoding



Operation

```
TEMP=PSR;
if (PSR[15] == 1)
   Saved_USP=R6 and R6=Saved_SSP;
   PSR[15]=0;
Push TEMP,PC† on the system stack
PC=mem[ZEXT(trapvect8)];
```

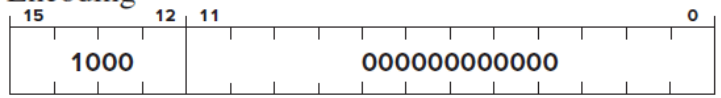
RTI

Return from Trap or Interrupt

Assembler Format

RTI

Encoding



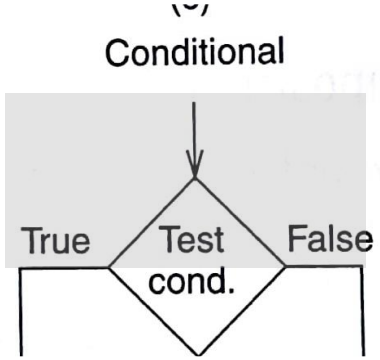
Operation

```
if (PSR[15] == 1)
   Initiate a privilege mode exception;
else
   PC=mem[R6]; R6 is the SSP, PC is restored
   R6=R6+1;
   TEMP=mem[R6];
   R6=R6+1; system stack completes POP before saved PSR is restored
   PSR=TEMP; PSR is restored
   if (PSR[15] == 1)
      Saved_SSP=R6 and R6=Saved_USP;
```

Assembly Programming

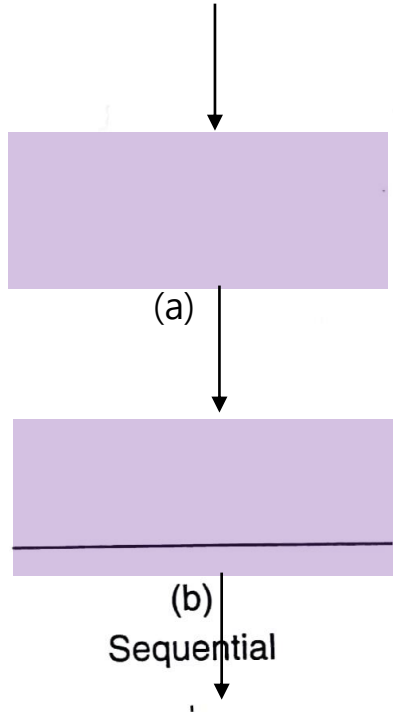
Programming Constructs

- Programming requires dividing a task, i.e., a unit of work into smaller units of work
- The goal is to replace the units of work with programming constructs that represent that part of the task
- There are three basic programming constructs
  - Sequential construct
  - Conditional construct
  - Iterative construct



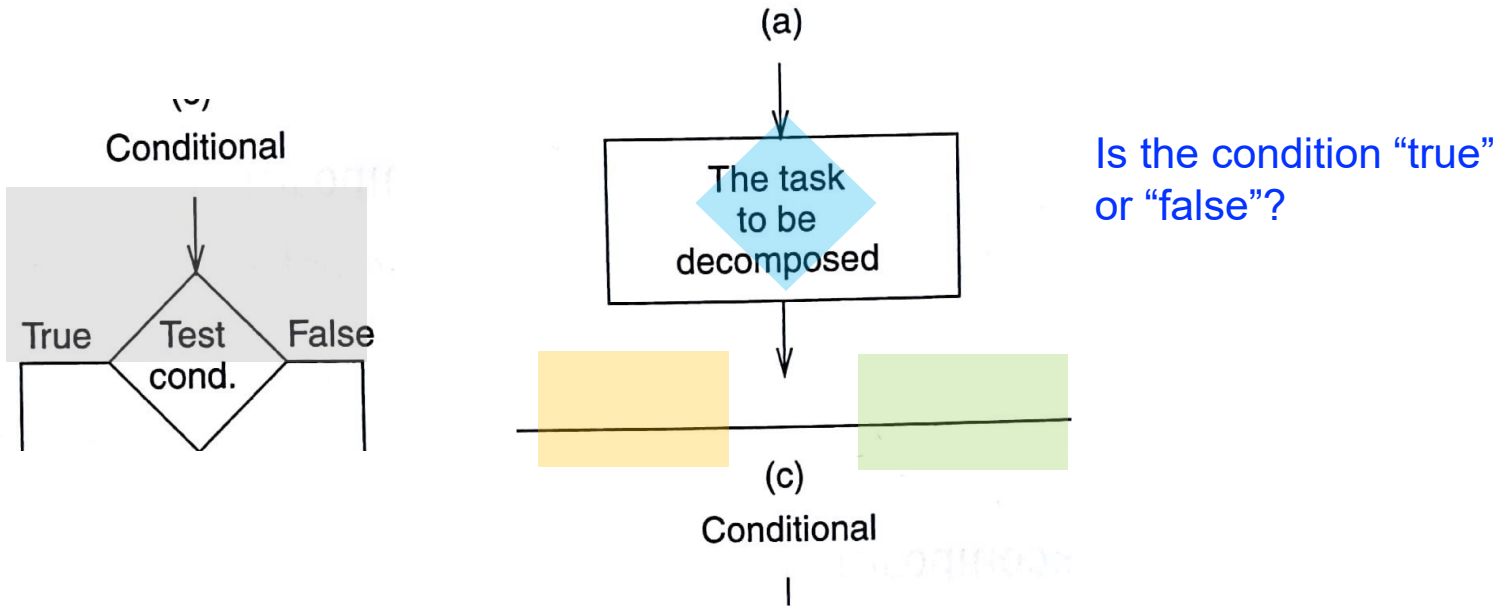
Sequential Construct

- The sequential construct is used if the designated task can be broken down into two subtasks, one following the other



Conditional Construct

- The conditional construct is used if the designated task consists of doing one of two subtasks, but not both

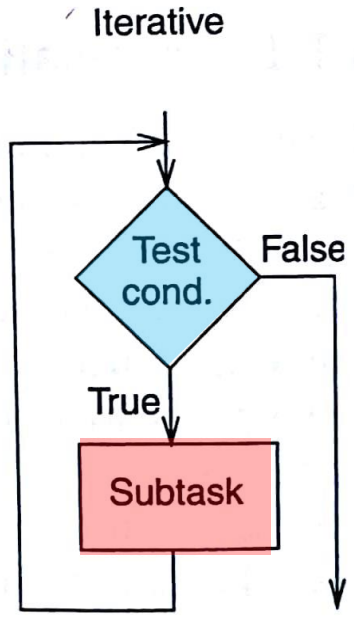


Is the condition "true" or "false"?

  - Either subtask may be "do nothing"
  - After the correct subtask is completed, the program moves onward
- E.g., if-else statement, switch-case statement

Iterative Construct

- The iterative construct is used if the designated task consists of doing a subtask a number of times, but only as long as some condition is true



Is the condition still “true”?

- E.g., for loop, while loop, do-while loop

Constructs in an Example Program

- Let us see how to use the programming constructs in an example program
- The example program counts the number of occurrences of a character in a text file
- It uses sequential, conditional, and iterative constructs
- We will see how to write conditional and iterative constructs with conditional branches

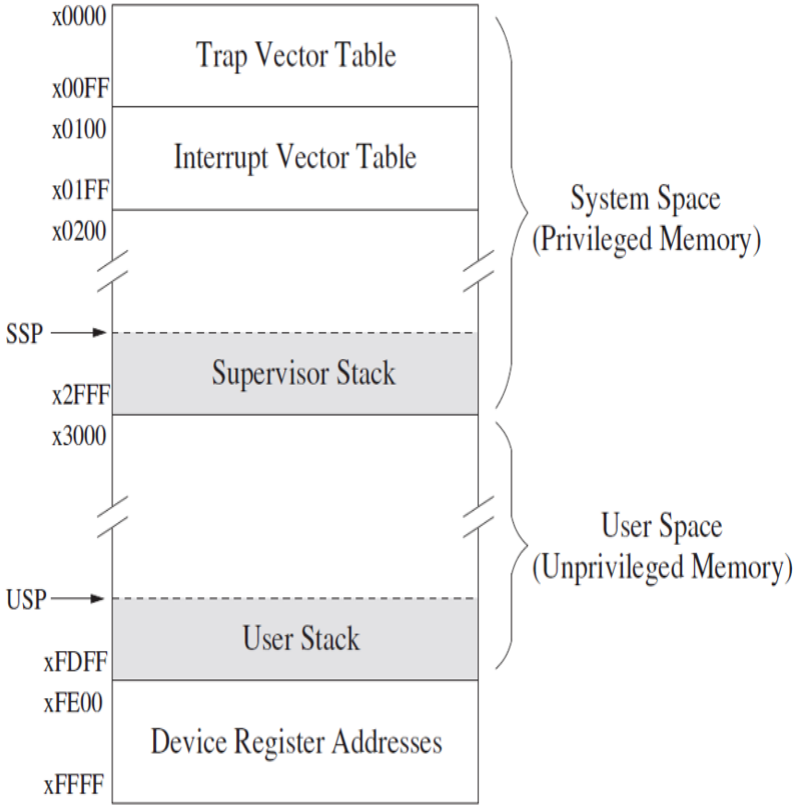
First LC-3 Program: Use of Conditional Branches for Looping

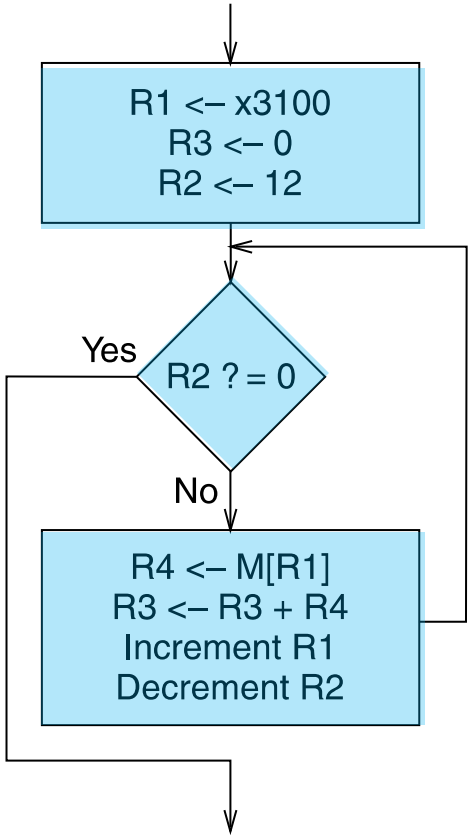
Readings

- *Introduction to Computing Systems: From Bits and Gates to C and Beyond by* Patt & Patel
  - Chapter 5, 6
  - Appendix A – LC3 ISA

An Algorithm for Adding Integers

- We want to write a program that adds 12 integers
  - They are stored in addresses 0x3100 to 0x310B
  - Let us take a look at the flowchart of the algorithm





R1: initial address of integers

R3: final result of addition

R2: number of integers left to be added

Check if R2 becomes 0 (done with all integers?)

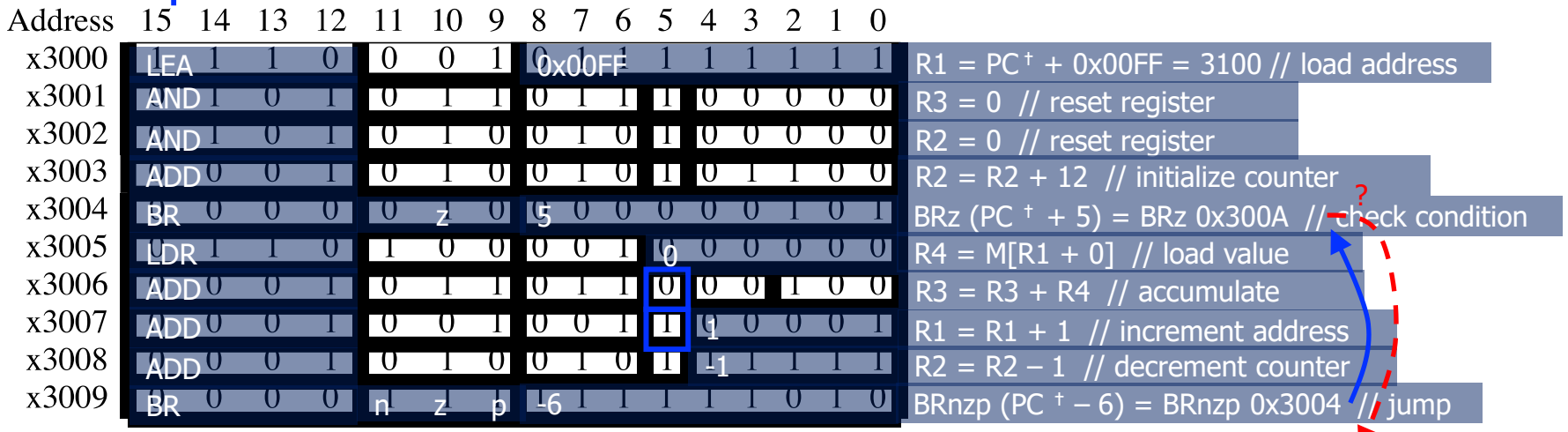
Load integer in R4

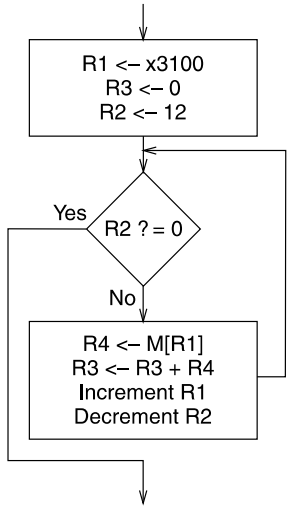
Accumulate integer value in R3

Increment address R1
Decrement R2

A Program for Adding Integers in LC-3

- We use conditional branch instructions to create a loop



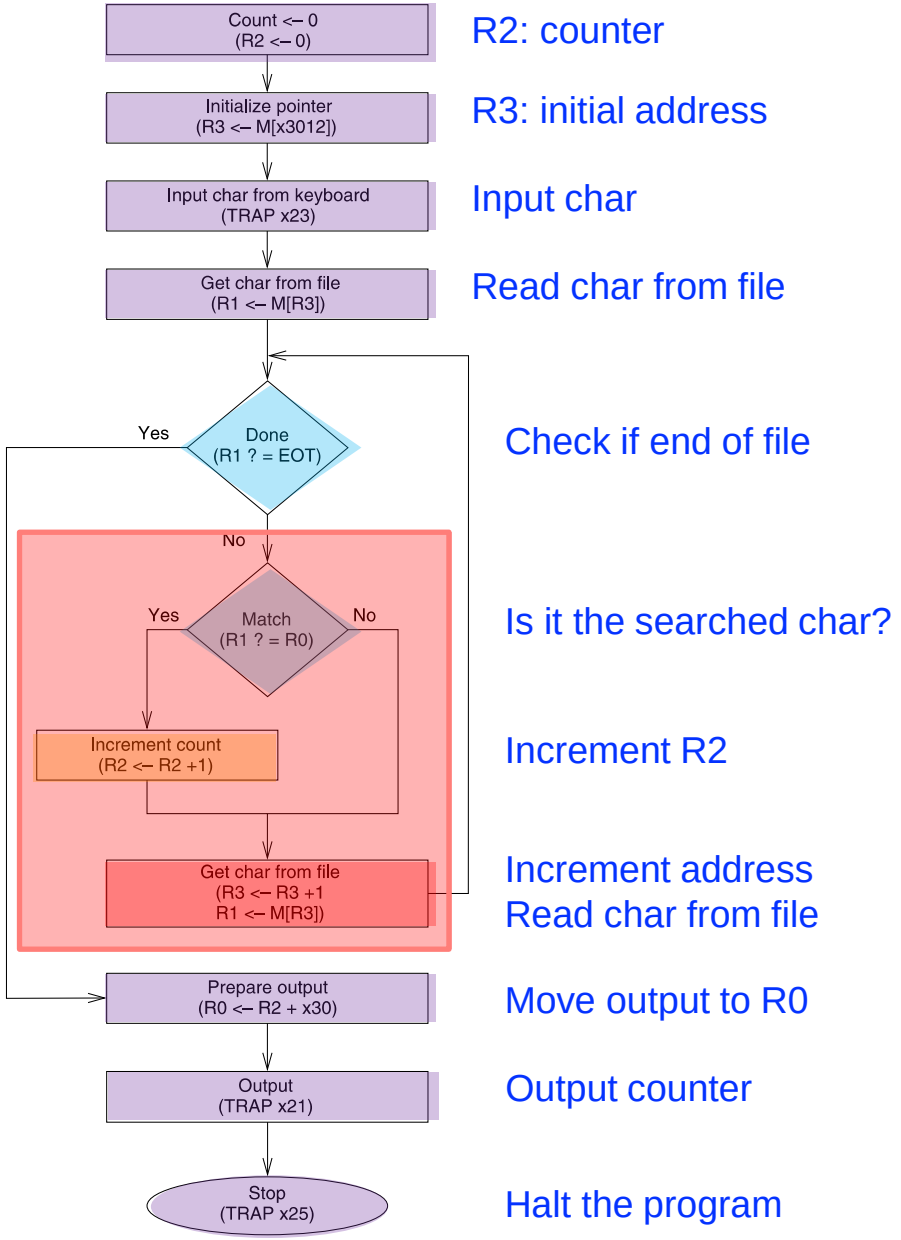


Bit 5 to differentiate both ADD instructions

† This is the incremented PC

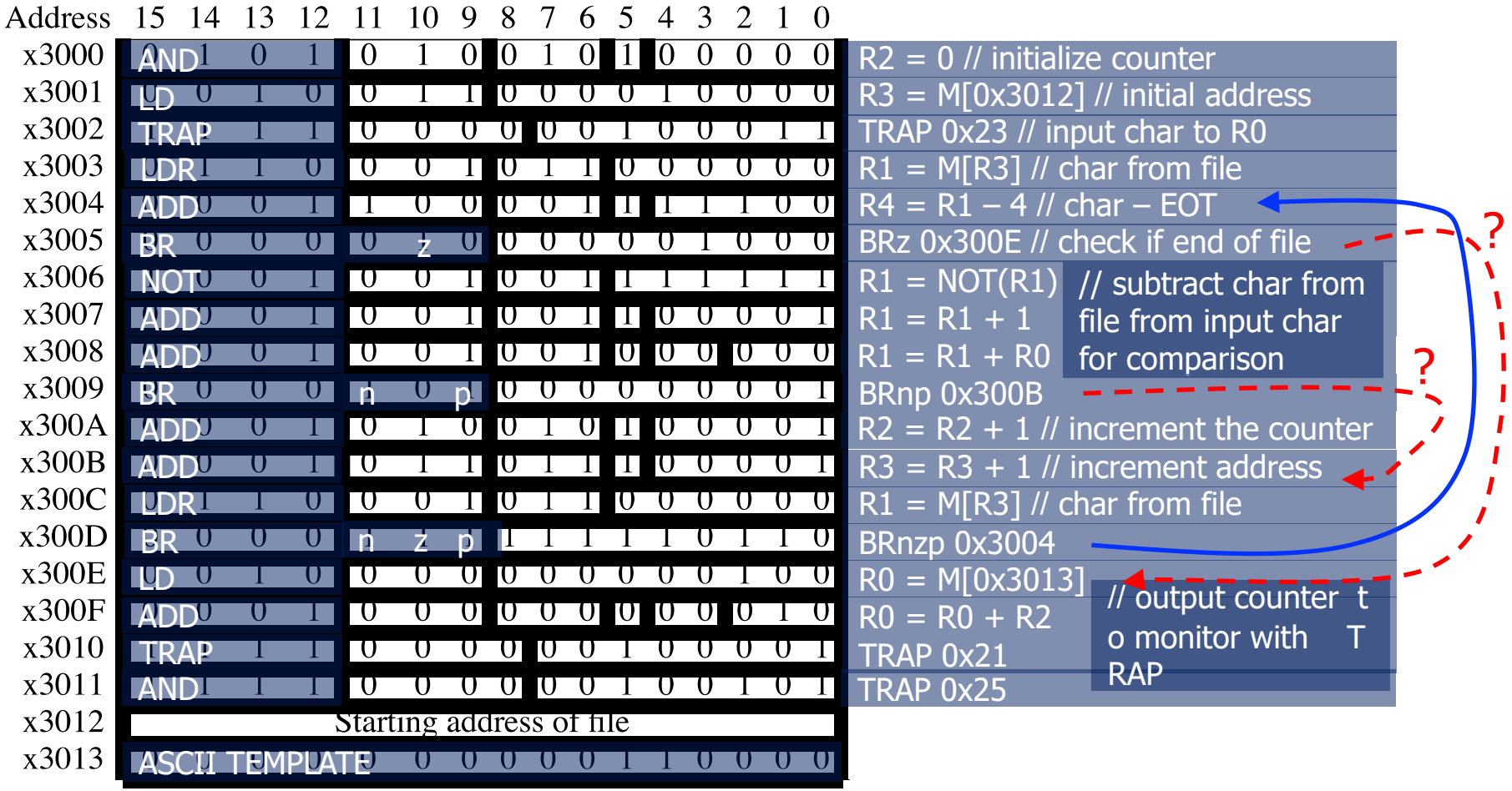
Counting Occurrences of a Character

- We want to write a program that counts the occurrences of a character in a file
  - Character from the keyboard (TRAP instr.)
  - The file finishes with the character EOT (End Of Text)
    - That is called a sentinel
    - In this example, EOT = 4
  - Result to the monitor (TRAP instr.)



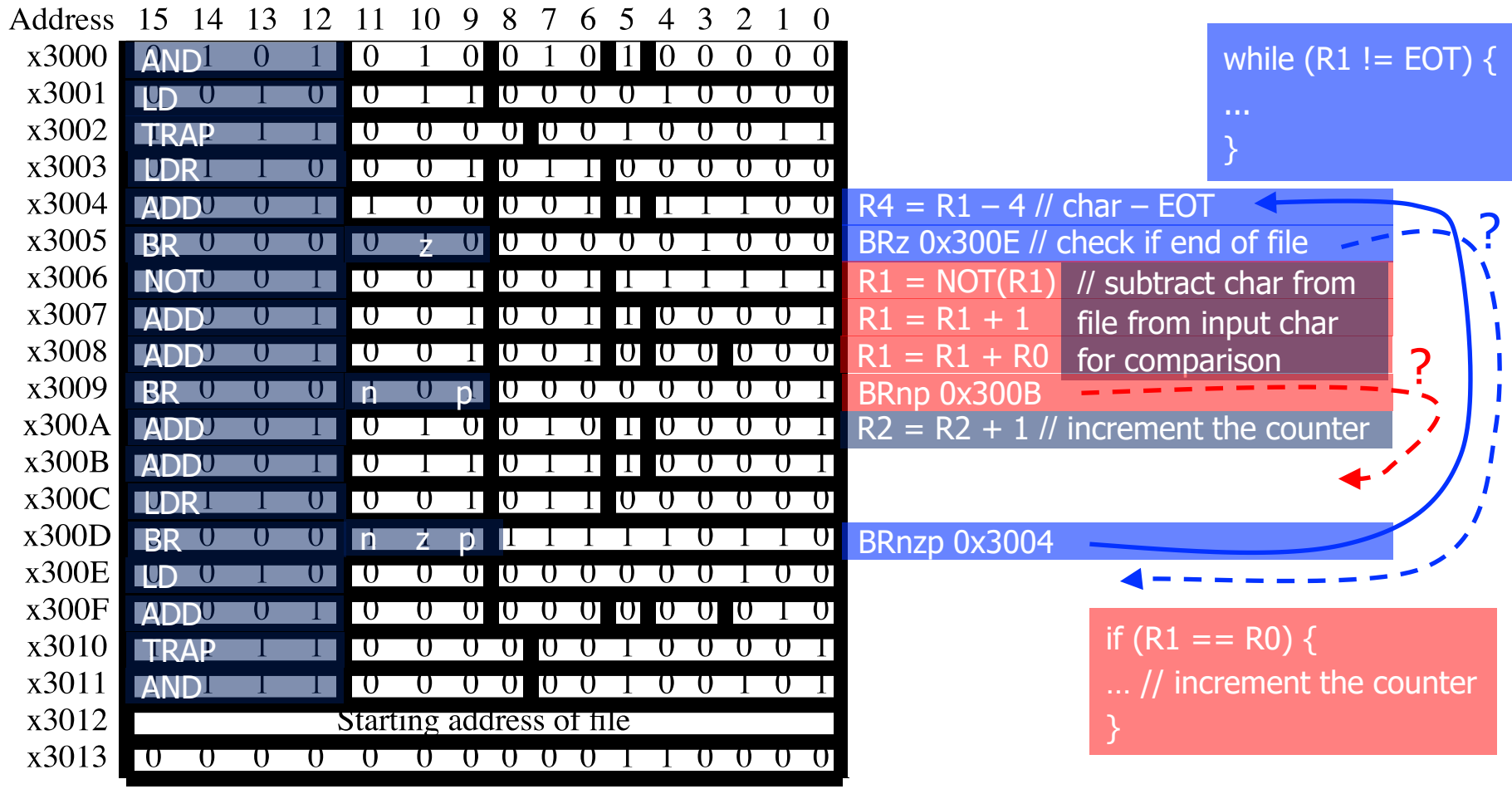
Counting Occurrences of a Char in LC-3

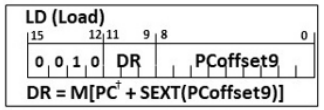
- We use conditional branch instructions to create a loops and if statements

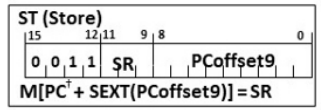


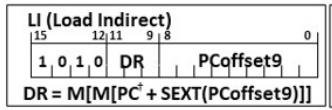
Programming Constructs in LC-3

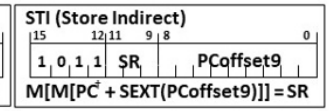
- Let us do some reverse engineering to identify conditional constructs and iterative constructs

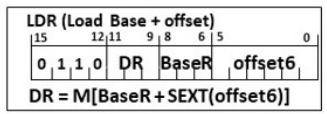


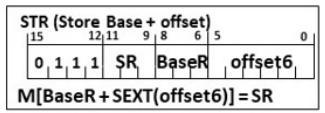


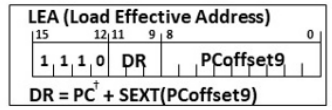


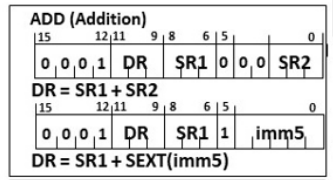


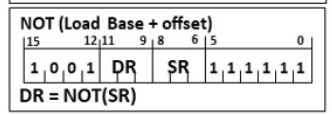


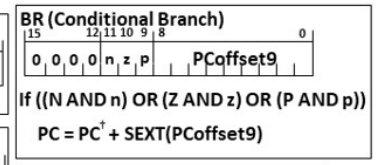


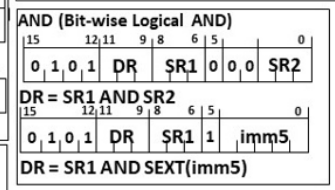


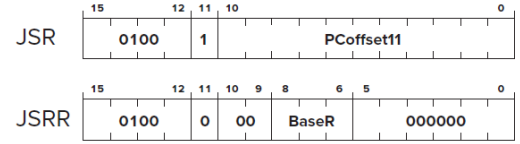


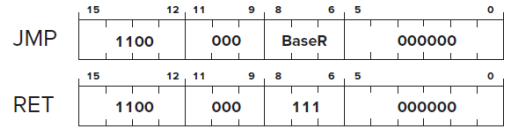


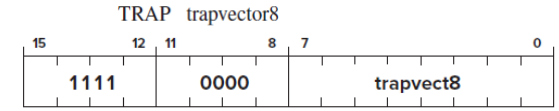












❑trapvect8 = service call

- 0x23 = Input a character from the keyboard
- 0x21 = Output a character to the monitor
- 0x25 = Halt the program

x3000																
x3001																
x3002																
x3003																
x3004																
x3005																
x3006																
X3007																
X3008																
X3009																
X300A																
X300B																
X300C																
X300D																
X300E																
X300F																
X3010																